EV Battery Course — Module 5: Battery Cell Formats and Internal Construction

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The same chemistry can be packaged as a cylinder, a prismatic can or a pouch, and the choice changes pack cost, cooling, safety, assembly and serviceability as much as chemistry does. This module opens each format and then looks at how cells are integrated into packs and vehicle bodies.

1. Learning objectives

- Describe the internal construction of cylindrical, prismatic and pouch cells.
- Explain wound (jelly-roll) vs stacked electrodes, and tabbed vs tabless designs.
- Explain cell safety devices, venting, swelling and gas generation.
- Explain why cells must be matched and balanced.
- Compare cell-to-module, cell-to-pack, cell-to-body and structural concepts on manufacturing, repair, structure, safety and service.

2. Fundamentals: the electrode sandwich

Every Li-ion cell, whatever its shape, is a long sandwich repeated many times:

copper foil → anode coating → separator → cathode coating → aluminium foil → cathode coating → separator → anode coating → copper foil …

Each layer is thin: foils about 6–15 μm, coatings about 50–100 μm, separator about 10–25 μm. A single EV cell can contain several square metres of electrode. The format is simply how that sandwich is folded and packaged.

- **Wound (jelly roll):** two long electrode strips and two separators are wound around a mandrel. Fast and cheap; used in cylindrical and many prismatic cells.
- **Stacked:** electrodes are cut into sheets and stacked (Z-folded separator or individual sheets). Better use of space in flat cells, more uniform pressure, but slower; used in pouch and some prismatic cells (e.g. BYD Blade).

3. The three formats

Feature	Cylindrical	Prismatic	Pouch
Casing	Steel (sometimes aluminium) can	Aluminium (or steel) rectangular can	Aluminium-laminated polymer film
Electrode assembly	Wound jelly roll	Wound flat roll or stacked	Stacked (sometimes folded)
Typical sizes	18650, 2170 (21 × 70 mm), 4680/46-series (46 mm dia.)	50–300+ Ah; Blade-type up to ~1 m long	50–100+ Ah, large flat sheets
Mechanical strength	High; can resists internal pressure	Medium-high	Low; needs external support
Safety devices	CID, PTC, vent in cap	Vent plate, sometimes fuse or overcharge device	None built in; seal opens under pressure
Swelling	Contained by can	Can bulges; needs end-plate compression	Large; needs controlled compression
Heat removal	Hard from the core; cooled from sides or bottom	Good through large faces or base	Very good through large faces
Packing efficiency	Gaps between cylinders	Very high	High, but needs frames
Cells per pack	Hundreds to thousands	~100–200	~100–300
Manufacturing	Very fast, highly automated, mature	Medium speed; larger parts	Flexible shape; sealing quality critical
Users	Tesla, Lucid, BMW (Neue Klasse), most 2W makers	CATL, BYD, Tata, most Chinese OEMs	LG, SK On, GM Ultium, Hyundai

4. Diagram: cross-sections

```
flowchart TB
  subgraph CYL["Cylindrical"]
    c1["Steel can (negative)"] --> c2["Jelly roll wound around a mandrel"] --
> c3["Cap: CID + PTC + vent (positive)"]
  end
  subgraph PRI["Prismatic"]
    p1["Aluminium can"] --> p2["Flat wound roll or stack"] --> p3["Lid:
terminals + vent plate + fill hole"]
  end
  subgraph POU["Pouch"]
    u1["Laminated film (PP / Al / nylon)"] --> u2["Stacked electrode sheets"]
--> u3["Tabs through heat-sealed edge"]
  end
```

5. Step-by-step: inside each cell

Cylindrical (e.g. 2170).

1. Anode and cathode strips are wound with separators into a jelly roll.
2. Tabs (or, in tabless designs, the folded foil edges) connect to the can bottom (negative) and the cap (positive).
3. Electrolyte is filled and the cap is crimped on.
4. Safety: the **CID (current interrupt device)** breaks the circuit when internal pressure rises; the **PTC** element raises resistance when hot; the **vent** releases gas.
5. **Tabless (e.g. 4680):** current flows along the whole foil edge instead of one tab, cutting the electron path length and internal resistance, which makes larger-diameter cells practical.

Prismatic.

1. A flattened roll or stack sits in an aluminium can, insulated by a film.
2. Tabs are welded to terminals on the lid.
3. The lid is laser-welded on; electrolyte is filled through a hole, later sealed.
4. A burst vent opens at a set pressure to release gas in a controlled direction.

Pouch.

1. Sheets are stacked with separator and tabs welded to lead tabs.
2. The stack is placed in a formed laminate pocket, electrolyte added, and the edges heat-sealed.
3. A gas pocket captures formation gas; it is cut off and resealed (degassing).

4. No rigid case, so the module must provide compression and protection.

6. Swelling, gas and why compression matters

- **Reversible swelling:** graphite expands about 10% when fully lithiated; silicon far more. Cells "breathe" with each cycle.
- **Irreversible swelling:** SEI growth and gas build-up over life thicken cells.
- **Gas generation:** mostly during formation (removed by degassing); later from electrolyte decomposition at high temperature or high voltage.
- **Compression:** moderate, uniform pressure keeps electrode layers in contact, improves life and limits gas pockets. Too much pressure damages the separator; too little lets layers separate. Modules use end plates, straps or foam pads to control it.

7. Cell balancing and matching

No two cells are identical: capacity, internal resistance and self-discharge vary. In a series string the lowest-capacity cell limits how far the pack can be charged and discharged. Factories grade cells into bins (e.g. capacity within ±0.5%, resistance within a few percent) and assemble packs from matched bins. Over life, the BMS balances them (Module 7).

8. Integration: from cell to vehicle

Concept	What it is	Advantages	Disadvantages	Examples
Cell-to-module (CTM)	Cells in modules; modules in pack	Repairable at module level; standard parts	Lower volume use (often ~40–50%); more parts	Most EVs before ~2020, many current EVs
Cell-to-pack (CTP)	Cells go directly into the pack	Fewer parts, ~15–30% better volume use, lower cost	Harder to repair; whole-pack thermal design critical	CATL CTP (Qilin), BYD Blade

Concept	What it is	Advantages	Disadvantages	Examples
Cell-to-body (CTB) / cell-to-chassis (CTC)	Pack lid is the vehicle floor; cells carry structural load	Lower mass, more energy in the same floor, stiffer body	Repair and second life very hard; crash design integrated	BYD CTB (Seal), Tesla structural pack (Model Y with 4680), Leapmotor CTC
Structural battery (materials)	Electrode materials themselves carry load	Ultimate mass saving	Research stage	Laboratory

9. Engineering equations

Cylinder volume:

$$V_{cyl} = \pi r^2 h$$

Ideal packing fraction of equal cylinders:

$$\phi_{square} = \frac{\pi}{4} \approx 0.785 \qquad \phi_{hexagonal} = \frac{\pi}{2\sqrt{3}} \approx 0.907$$

Compression force from pressure:

$$F = p \times A$$

Pack volume utilisation:

$$\eta_{vol} = \frac{\sum V_{cells}}{V_{pack}}$$

10. Worked examples

Example 1: 4680 vs 2170 cell volume.

- 4680: π × 23² × 80 = **132,951 mm³ (0.133 L)**
- 2170: π × 10.5² × 70 = **24,245 mm³ (0.024 L)**
- Ratio about **5.5×**. Fewer, bigger cells mean fewer welds and less handling.

Example 2: cell count and welds for 75 kWh. Assume about 85 Wh per 4680 and about 18 Wh per 2170 (illustrative values).

- 4680: 75,000 ÷ 85 ≈ **882 cells** → about 1,764 cell-to-busbar welds (2 per cell)
- 2170: 75,000 ÷ 18 ≈ **4,167 cells** → about 8,334 welds

Example 3: packing. In hexagonal packing, cylinders fill at most 90.7% of a layer's area; square packing fills 78.5%. Prismatic cells can approach 100% before allowing for gaps, cooling and barriers.

Example 4: compression force on a pouch cell. Face 100 mm × 300 mm = 0.03 m²; target pressure 100 kPa.

- F = 100,000 × 0.03 = **3,000 N** per cell face. End plates for a 20-cell module must hold this force over life without creeping.

Example 5: stack growth. Each 10 mm cell grows 5% (0.5 mm) by end of life; 20 cells in a stack: **10 mm** total. The module must accommodate this without losing compression.

11. Real-world examples

- **Tesla:** 18650 (Model S/X), 2170 (Model 3/Y), 4680 in Cybertruck and some Model Y; also LFP prismatic in standard-range models.
- **BYD Blade:** long, thin LFP prismatic cells that act as structural beams in a CTP/CTB pack.
- **CATL Qilin:** CTP pack with cooling plates between cells on their large faces.
- **GM Ultium / Hyundai E-GMP:** large pouch cells in modules.
- **India:** Tata EVs use prismatic LFP packs; most electric two-wheelers use cylindrical cells (18650/21700) or small prismatic packs; Ola Electric has built 4680-format cells.

12. Manufacturing applications

- **Throughput:** cylindrical winding and assembly lines run at very high speed (hundreds of cells per minute); prismatic and pouch lines are slower per cell but handle more Ah per cell.
- **Critical welds:** tab-to-terminal (ultrasonic or laser), lid-to-can laser seam (prismatic), heat seal (pouch). Each is a leak and resistance risk.
- **Alignment:** anode overhang of about 0.5–2 mm beyond the cathode on all sides is critical to prevent lithium plating at edges; vision systems check it.
- **Repairability vs cost:** CTP and CTB lower BOM cost but turn a single-cell fault into a whole-pack replacement, which shifts warranty cost and affects insurance and resale.

13. Common misconceptions

- **"Cylindrical is old-fashioned."** The 46-series cylindrical is a growing next-gen format.
- **"Pouch cells are less safe."** Safety depends on pack design and chemistry; pouches simply move protection to the module.
- **"Bigger cells are always better."** They reduce part count but make heat removal from the core harder and raise the energy released in a single-cell failure.
- **"CTP means no structure."** CTP removes modules, not structure; the cells or pack shell carry load.

14. Practical case study: pouch-cell swelling in a module

Problem. A fleet of electric buses shows modules with cracked plastic frames after 2 years, and cells with capacity fade above expectations.

Root cause. Foam compression pads took a permanent set at high temperature, so pressure dropped; cells swelled unevenly, gas pockets formed and layers delaminated. Frames cracked from uneven loading.

Corrective action. Change to a foam rated for compression set at 60 °C, add steel end plates with a defined preload, and add a swelling check (cell thickness) in periodic service.

Lesson. Mechanical design of compression is part of battery life engineering.

15. Revision notes

- Cylindrical = wound, strong can, built-in CID/PTC/vent, many cells.
- Prismatic = large Ah, aluminium can, vent plate, good packing.
- Pouch = stacked, light, flexible shape, needs external compression.
- Tabless designs cut internal resistance in large cylinders.
- CTM → CTP → CTB trades repairability for cost, mass and volume.

16. Glossary

Term	Meaning
Jelly roll	Wound electrode and separator assembly
CID	Current interrupt device; opens under internal pressure

Term	Meaning
PTC	Positive temperature coefficient element; resistance rises with heat
Tabless	Current collected along the whole foil edge
Overhang	Anode extending beyond cathode edges
Degassing	Removing formation gas from a pouch cell
CTP / CTB / CTC	Cell-to-pack / cell-to-body / cell-to-chassis

17. Knowledge check

Conceptual

1. Why is the anode made slightly larger than the cathode?
2. Why did larger cylindrical cells need tabless designs?
3. What does a CID do, and why are pouch cells without one?
4. Why do pouch modules need compression?
5. What are the service consequences of cell-to-body integration?

Numerical

1. Compute the volume of an 18650 cell (18 mm × 65 mm).
2. A pack needs 60 kWh using 300 Wh prismatic cells. How many cells? If arranged in 96 series groups, how many in parallel?
3. A prismatic cell face is 0.15 m × 0.1 m; target preload 150 kPa. What force is needed?

Application

1. A two-wheeler start-up must choose 21700 cylindrical or 100 Ah prismatic LFP cells. List the trade-offs for assembly, cooling, safety and cost.
2. An insurer asks about repair costs of CTB vs CTM. What would you tell them?

18. Answer key

Conceptual

1. So every lithium ion leaving the cathode edge has graphite to go into; otherwise lithium plates at the anode edge.

2. In a single-tab design, current travels the full length of the foil, so resistance and heat rise with cell size. Tabless designs collect current along the whole edge, shortening the path.
3. It mechanically disconnects the cell when internal pressure rises from overcharge or gassing. Pouch cells have no rigid case to mount it against; protection moves to the BMS and module.
4. To keep electrode layers in contact, prevent gas pockets and control swelling; this improves life and consistency.
5. Cells can't be reached without disturbing the body structure; a cell fault may mean replacing a large structural assembly; second-life reuse is hard; crash damage can write off the vehicle.

Numerical

1. $\pi \times 9^2 \times 65 = \pi \times 81 \times 65 =$ **16,540 mm³ (16.5 cm³)**.
2. 60,000 ÷ 300 = **200 cells**. 200 ÷ 96 = 2.08, which isn't whole. Choose 100S2P = 200 cells (if V and Ah suit) or 96S2P = 192 cells for 57.6 kWh. Pack design rounds to whole numbers (Module 6).
3. A = 0.015 m²; F = 150,000 × 0.015 = **2,250 N**.

Application

1. 21700: high automation, mature supply, small failure energy per cell, but many welds and harder cooling. 100 Ah prismatic: few cells and welds, simpler BMS harness, easier cooling through faces, but a single-cell failure releases more energy and the format is less standard among 2W suppliers. Cost per kWh is often similar at cell level; prismatic usually wins at pack-assembly level.
2. CTM: faulty modules can often be replaced. CTB: cells are integrated with the floor; repairs may require replacing the pack-body assembly, so repair cost is high and minor floor damage can be a total loss. Diagnostic data and OEM repair procedures matter.

19. Further learning

- Battery University BU-301a, "Types of Battery Cells."
- Tesla Battery Day presentation (2020): tabless 4680 rationale.
- CATL and BYD technical presentations on CTP and Blade.
- Korthauer, *Lithium-Ion Batteries: Basics and Applications*: cell design chapters.

Next: Module 6, pack design and the 60 kWh pack project.